

Mohamed Shamik

Software Engineering Undergraduate

mdshamik@outlook.com | +65 8262 3134 | GitHub: <https://github.com/Mohdshamik11> | LinkedIn: <https://www.linkedin.com/in/mohamedshamik/>

SUMMARY

Software Engineering undergraduate with hands-on experience across full-stack development and IT infrastructure, gained through internships and academic projects. Proficient in Python, SQL, and cloud/DevOps tooling (Docker, AWS, CI/CD). Actively learning and building AI/ML skills, ML workflows, NLP, and LLM/computer vision projects.

SKILLS

Languages: Python, SQL
Frameworks: PyTorch, FastAPI, Flask
Cloud & DevOps: AWS (EC2), Vercel, GitHub Actions (CI/CD), Nginx
Tools & Platforms: Docker, GitHub, VSCode
Databases: MariaDB, Supabase, Firebase, SQLite

EDUCATION

B.Eng in Software Engineering (Hons) Singapore Institute of Technology	Sep 2024 – Ongoing
Diploma in Infocomm & Security Nanyang Polytechnic	Apr 2019 – May 2022

PROFESSIONAL EXPERIENCE

Micro2000 IT Installer	May 2025 – Aug 2025
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- Executed device provisioning, sanitisation, and assessment for the Schools Standard ICT Operating Environment (SSOE) project.
- Managed small team operations and liaised with school staff during handovers and takeovers.
- Independently handled devices, documentation, and on-site project tasks, demonstrating accountability.
- Contributed to a smooth SSOE rollout, ensuring 100% compliance with project standards.

Zicom Holdings Pte Ltd IT & Web Support Intern	Dec 2021 – Apr 2022
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- Revamped the company's official website using a CMS, updating product and service listings to keep content current and improve customer-facing presentation.
- Consulted with the IT Manager to assess logistics and hardware upgrade needs, coordinating procurement of necessary equipment.
- Delivered IT Helpdesk support across hardware, software, and network connectivity issues for the IT and Engineering departments.
- Conducted a virtual training session for a team of 5, sharing networking fundamentals to strengthen the team's technical knowledge base.

ACADEMIC PROJECTS

Bachelor of Engineering in Software Engineering - Singapore Institute of Technology

1. HitsV2 - Ongoing
 - Building a full-stack smart-campus data portal (React 19, FastAPI, Databricks) enabling self-service discovery, preview, and governed access to IoT datasets across 4 domains.
 - Designed a normalised 7-table access-control schema and implemented a request/approve/grant/revoke workflow with owner approval inboxes and notifications.
 - Architected the system so storage can swap from in-memory to Lakebase Postgres without API or UI changes.
 - Building email/password authentication for the portal using FastAPI and bcrypt password hashing, with server-side sessions delivering via Secure, httpOnly, SameSite cookies to mitigate XSS and CSRF.
 - Enforced role-based authorization (user/owner/admin/custodian) across preview, download, and access-request routes through a centralized identity layer, and added a React login flow with protected routes.
2. Full Stack Application with Security Measures
 - Built a web app for personal finance tracking with consent-based data sharing between individuals, households, and financial advisors.
 - Built a CI/CD pipeline in GitHub Actions, automating tests, linting, and type-checks on every PR, with zero-touch deployments to AWS EC2 on merge.
 - Implemented authentication security including bcrypt password hashing and per-IP rate limiting to prevent brute-force attacks.
 - Designed a token-based email invitation system enabling users to onboard household members and advisors into scoped roles.
3. Assisted Remote-Controlled Ball Rover
 - Built a full-stack IoT control system for a remote-controlled rover, pairing embedded C firmware with a browser dashboard over real-time MQTT/Wi-Fi communication
 - Implemented sensor fusion of mmWave radar and IMU data for autonomous obstacle detection, with deterministic arbitration between manual and safety-override control
 - Developed a 360° autonomous scanning feature that samples obstacle data across 8 directions and renders a live ASCII radar map in the web UI

Diploma in Infocomm & Security

1. Network Design and Implementation
 - Designed and configured a multi-router, multi-switch network topology, applying subnetting and routing protocols to ensure reliable connectivity across segments.
 - Implemented network security controls including firewalls, Intrusion Prevention Systems (IPS), VPN tunnels, and traffic encryption to safeguard against unauthorized access.
 - Configured user authentication and role-based access controls, restricting server access to authorized personnel only.

PERSONAL PROJECTS

1. Image Quality Classifier & Enhancer
 - Built a convolutional neural network (CNN) that automatically detects five common photo problems in one pass: blur, over-exposure, under-exposure, noise, and low contrast.
 - Generated a synthetic training set from a pool of clean photos. For each image, code produces the five defect types at randomized strengths, along with multi-defect combinations and untouched copies, and creates the labels automatically from the defects applied, so no manual annotation was needed.
 - Built a second model, a U-Net trained with an adversarial (GAN) loss to repair blur, specifically a discriminator network judged real vs. repaired image crops during training, pushing the generator toward genuinely sharp output instead of a safe, blurry average.
 - Repairs flagged for exposure, contrast and noise are done by classical enhancements to the image.

- Deployed the whole tool as a live web app built with Streamlit, where a user uploads a photo and gets the defect flags plus an enhanced version.

CERTIFICATIONS

- AWS Certified Cloud Practitioner — In Progress